

JESS Thermodynamic Database v8.9

Einsteinium

24-Jan-23

Reaction No. 14862							
DTPA-5 + Es+3 = Es+3_DTPA-5							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NH4ClO4	lgK:22.6(0.07SD)	4	MIX	999

Reaction No. 53018							
Cl-1 + Es+3 = Es+3_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:-0.02(1SF)	4	MDS	5398

Reaction No. 53329							
SO4-2 + Es+3 = Es+3_SO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.81(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:3.81(3SF)	4	MDS	5398

Reaction No. 53330							
2<SO4-2> + Es+3 = Es+3_SO4-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.66(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:5.66(3SF)	4	MDS	5398

Reaction No. 53331							
3<SO4-2> + Es+3 = Es+3_SO4-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.07(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:5.07(3SF)	4	MDS	5398

Reaction No. 55641							
Es+3 + e-1 = Es+2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-1.3(2SF)	3	CRV	6330
2	25	0	Inf. Dilution	E:-1.3(2SF)	3	ENS	7276

3	25	0	Inf. Dilution	E:-1.3(2SF)	3	CRV	7761
4	25	0	Inf. Dilution	E:-1.3(2SF)	4	CRV	22519
5	25	0	Inf. Dilution	E:-1.5(2SF)	0	CRV	22551
6	25	0	Unknown	E:-1.4(0.1SD)	0	ENS	5256
7	25	0	Inf. Dilution	dH:171.5(4SF) kJ	3	CRV	7761

Reaction No. 58017							
SCN-1 + Es+3 = Es+3_SCN-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.498(4SF)	1	EOR	
2	25	1	NaClO4	lgK:0.6(0.03SD)	5	MSE	5612

Reaction No. 66177							
CDTA-4 + Es+3 = Es+3_CDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:20.1(3SF)	4	CRV	552

Reaction No. 69871							
Es+3_Citric-3(2) + 6<H+1> = Es+3 + 2<H+1(3)_Citric-3>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:14.09(4SF)	5	MDS	2330

Reaction No. 69872							
Es+3_H+1_Citric-3(2) + 5<H+1> = Es+3 + 2<H+1(3)_Citric-3>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:9.68(3SF)	5	MDS	2330

Reaction No. 69878							
Es+3_H+1_Citric-3 + Citric-3 = Es+3_H+1_Citric-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4	lgK:10.64(4SF)	3	MSE	2330

Reaction No. 69879							
2<Citric-3> + Es+3 = Es+3_Citric-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.76(4SF)	1	EOR	
2	25	0.1	NaClO4	lgK:12.07(4SF)	3	MSE	2330

Reaction No. 70269							
$\text{Es}+3 + 3\langle\text{e}-1\rangle = \text{Es}(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -97.4 (3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$E: -1.91 (3\text{SF})$	0	CRV	6330
3	25	0	Inf. Dilution	$E: -1.98 (3\text{SF})$	0	ENS	7276
4	25	0	Inf. Dilution	$E: -1.91 (3\text{SF})$	0	CRV	7761
5	25	0	Inf. Dilution	$E: -1.91 (3\text{SF})$	0	CRV	22519
6	25	0	Inf. Dilution	$E: -1.98 (3\text{SF})$	0	CRV	22551
7	25	0	Inf. Dilution	$\text{dH}: 584.8 (4\text{SF}) \text{ kJ}$	3	CRV	7761

Reaction No. 70272							
$\text{Es}+2 + 2\langle\text{e}-1\rangle = \text{Es}(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$E: -2.23 (3\text{SF})$	4	CRV	6330
2	25	0	Inf. Dilution	$E: -2.3 (2\text{SF})$	0	ENS	7276
3	25	0	Inf. Dilution	$E: -2.23 (3\text{SF})$	3	CRV	7761
4	25	0	Inf. Dilution	$E: -2.23 (3\text{SF})$	4	CRV	22519
5	25	0	Inf. Dilution	$\text{dH}: 413.1 (4\text{SF}) \text{ kJ}$	3	CRV	7761

Reaction No. 72050							
$\text{Es}+2_{\text{OH}-1(2)}(\text{s}) + 2\langle\text{e}-1\rangle = \text{Es}(\text{s}) + 2\langle\text{OH}-1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$E: -2.3 (2\text{SF})$	3	CRV	7761
2	25	0	Inf. Dilution	$\text{dH}: 386.3 (4\text{SF}) \text{ kJ}$	3	CRV	7761

Reaction No. 72051							
$\text{Es}+3_{\text{OH}-1(3)}(\text{am.}, \text{s}) + 3\langle\text{e}-1\rangle = \text{Es}(\text{s}) + 3\langle\text{OH}-1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$E: -2.42 (3\text{SF})$	3	CRV	7761
2	25	0	Inf. Dilution	$\text{dH}: 600. (3\text{SF}) \text{ kJ}$	3	EES	
Note (2): By analogy with the crystalline solid							

Reaction No. 72052							
$\text{Es}+3_{\text{OH}-1(3)}(\text{s}) + 3\langle\text{e}-1\rangle = \text{Es}(\text{s}) + 3\langle\text{OH}-1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$E: -2.42 (3\text{SF})$	3	CRV	7761

2	25	0	Inf. Dilution	dH:618.5(4SF)kJ	3	CRV	7761
---	----	---	---------------	-----------------	---	-----	------

Reaction No. 72053							
Es+3_OH-1(3)_(s) + e-1 = Es+2_OH-1(2)_(s) + OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-44.9(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-2.6(2SF)	0	CRV	7761
Note (2): Low wgt for inter-reaction consistency							
3	25	0	Inf. Dilution	dH:225.0(4SF)kJ	3	CRV	7761

Reaction No. 74056							
Es+4 + e-1 = Es+3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:4.5(2SF)	4	CRV	22551

Reaction No. 76138							
Es+3_SCN-1 + SCN-1 = Es+3_SCN-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:-1.96(3SF)	5	MDS	11516
Note (1): Harmon H et al., J. Inorg. Nucl. Chem., 1972, 34, 1381							

Reaction No. 79099							
2<Citric-3> + H+1 + Es+3 = Es+3_H+1_Citric-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.8(4SF)	1	ELC	

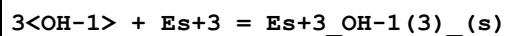
Reaction No. 79103							
Citric-3 + H+1 + Es+3 = Es+3_H+1_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.532(4SF)	1	ELC	

Reaction No. 79178							
2<OH-1> + Es+2 = Es+2_OH-1(2)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.367(4SF)	1	ELC	

Reaction No. 79179							
3<OH-1> + Es+3 = Es+3_OH-1(3)_(am.,s)							

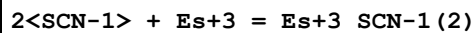
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.36(4SF)	1	ELC	

Reaction No. 79180



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.36(4SF)	1	ELC	

Reaction No. 79439



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.1375(4SF)	1	ELC	

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review
EES	Personal estimate
ELC	Linear Combination of Reactions

ENS	Not specified
EOR	Other Reaction Constants
MDS	Distribution between two phases
MIX	Ion exchange
MSE	Solvent Extraction

References

Reference No.: 552(95SMM)

NIST Critical Stability Constants of Metal Complexes Database, 1995 Smith RM; Martell AE; Motekaitis RJ; Version 2.0; National Institute of Standards and Technology, Gaithersburg, USA

Reference No.: 999(91MaM)

Estimation of Constants for JESS Database, 1991 May PM; May RF

Reference No.: 2330(74HHB)

J. Inorg. Nucl. Chem., 1974, 36, 2361 - 2366; Hubert S; Hussonnois M; Brillard L; Goby G; Guillaumont R; Determination Simultanee de Constantes de Formation de Complexes Critrique de l'Americium, du Curium, du Californium, de l'Einsteinium et du Fermium

Reference No.: 5256(91MiK)

Coord. Chem. Rev., 1991, 109, 1 - 5; Mikheev NB; Kamenskaya AN; Complex Formation of the Lanthanides and Actinides in Lower Oxidation States

Reference No.: 5398(82Hog)

IUPAC Chemical Data Series No. 21, 1982 Hogfeldt E; Stability Constants of Metal-ion Complexes, 2nd Suppl., Part A, Inorganic Ligands; Pergamon, Oxford

Reference No.: 5612(80KhM)

J. Inorg. Nucl. Chem., 1980, 42, 109 - 113; Khopkar PK; Mathur JN; Complexing of Californium(III) and other Trivalent Actinides by Inorganic Ligands

Reference No.: 6330(95LiF)

CRC Handbook of Chemistry and Physics, 76th Edn., 1995 Lide DR; Frederikse HPR; A Ready-Reference Book of Chemical and Physical Data; CRC Press, Cleveland, Ohio, U.S.A.

Reference No.: 7276(88CoW)

Advanced Inorganic Chemistry, 5th Edn., 1988 Cotton FA; Wilkinson G; John Wiley & Sons Inc., N.Y., U.S.A

Reference No.: 7761(89Bra)

J. Phys. Chem. Ref. Data, 1989, 18, 1 - 21; Bratsch SG; Standard Electrode Potentials and Temperature Coefficients in Water at 298.15 K

Reference No.: 11516(00PeP)

IUPAC Stability Constants Database, 2000 Pettit LD; Powell HKJ; Release 4; Academic Software, U.K.

Reference No.: 22519(05Lid)

CRC Handbook of Chemistry and Physics, 86th Edn., 2005 Lide DR; A Ready-Reference Book of Chemical and Physical Data; Taylor & Francis, Boca Raton, USA

Reference No.: 22551(06Inz)

Encyclopedia of Electrochemistry, Vol. 7a, 2006, 17 - 75; Inzelt G; Standard, Formal and Other Characteristic Potentials of Selected Electrode Reactions